COM5231 · Introduction to Wireless and Ad Hoc Networking

Answers to Your Photographed Papers

Test 2 2026 · Test 1 (A) · Test 1 (B) · Test 2 2025 · Exam page 4 (Q4e, Q5)

SUPPLEMENTARY content not found in the uploaded lecture notes

NOTES ERROR a mistake in the lecture notes, with the verified correction

EXAM how the topic is or could be examined

★ high-priority, frequently examined content

Exam preparation pack, built from the uploaded lecture materials and past papers

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About These Answers

These are model answers for the five pages you photographed, **in the same order as your photos**:

1. **COM5231 Test 2 2026** (50 marks)
2. **Test 1** (version A, 14 questions)
3. **COM5231 Test 1** (version B, 9 questions)
4. **Test 2 2025** (100 marks)
5. **Exam paper, page 4**: Question 4(e) and Question 5

Every answer is written out in full, so nothing refers you to another page. Marks are shown in brackets, and answer length follows the marks.

- Sources from your notes are in square brackets: [WC] wireless channel, [WCP] wireless problems, [MA] multiple access, [MX] multiplexing, [AT] access-technologies article, [SS] spread spectrum, [OF] OFDM, [ED] error worksheet, [WL] WLAN slides, [FZ] Forouzan WLAN chapter, [BT-B]/[BT-S] Bluetooth book/slides.
- **Routing answers (Test 2 2026 Q1 and Q9; Test 2 2025 Q1–Q3) are SUPPLEMENTARY.** Your notes don't cover routing, so these follow the standard textbook method.
- No memos were provided. These are **model answers**, not the lecturer's marking guide. Every calculation has been checked by computer.

COM5231 Test 2 2026 (50 marks)

Q1. Distinguish between routing, packet forwarding and switching. For each, state its purpose and the information used to make its decision. (6)

SUPPLEMENTARY (2 marks each).

- **Routing**: **purpose**: determine the **end-to-end path** that packets take from source to destination across the network (a network-wide, control-plane function performed by routing algorithms such as Dijkstra/link-state and Bellman-Ford/distance-vector). **Information used**: the **network topology and link costs**, learned by exchanging routing messages with other routers. The output populates the **routing/forwarding table**.
- **Packet forwarding**: **purpose**: inside one router, **move each arriving packet from its input port to the correct output port/link** (a local, per-packet, data-plane action). **Information used**: the packet's **destination IP (network-layer) address**, looked up in the **forwarding table** (longest-prefix match).
- **Switching**: **purpose**: move **frames** to the correct outgoing port **within a LAN** at the link layer (also the term for transferring packets across a router's internal switching fabric). **Information used**: the frame's **destination MAC address**, looked up in the **switch table**, which the switch builds by **self-learning** from the source MAC addresses of arriving frames.

Q2. Data byte 11001010; even-parity Hamming code; positions numbered from 1 at the left; check bits at 1, 2, 4, 8; data bits left to right in the remaining positions; no overall parity bit.

(a) Minimum number of check bits; show the inequality and verify. (2)

Inequality: **2^r ≥ m + r + 1**, with m = 8 data bits.

- r = 3: $2^3 = 8 \geq 8 + 3 + 1 = 12$? **No**.
- r = 4: $2^4 = 16 \geq 8 + 4 + 1 = 13$? **Yes**. **Minimum r = 4** check bits, giving a 12-bit codeword.

(b) Data-bit placement and value of each check bit, with the positions covered. (4)

Position	1	2	3	4	5	6	7	8	9	10	11	12
Bit	P1	P2	**1**	P4	**1**	**0**	**0**	P8	**1**	**0**	**1**	**0**

Check bit	Positions covered	Data bits	Number of 1s	Even-parity value
P1	1, 3, 5, 7, 9, 11	1, 1, 0, 1, 1	4	**0**
P2	2, 3, 6, 7, 10, 11	1, 0, 0, 0, 1	2	**0**
P4	4, 5, 6, 7, 12	1, 0, 0, 0	1	**1**
P8	8, 9, 10, 11, 12	1, 0, 1, 0	2	**0**

(c) Complete transmitted codeword (positions 1 → 12). (2)

0 0 1 1 1 0 0 0 1 0 1 0 → 001110001010

Q3. CRC: data 110101, generator 1011 ($G(x) = x^3 + x + 1$), modulo-2 division.

(a) Number of zeros appended and the resulting bit string. (1)

Degree 3 → **append 3 zeros** → **110101000**.

(b) Show the division and the CRC remainder. (3)

Step	Window	⊕ 1011	Result	Working value
1	1101	1011	0110	011001000
2	1100	1011	0111	001111000
3	1111	1011	0100	000100000
4	1000	1011	0011	000001100
5	1100	1011	0111	000000111

```
110101000
1011
----
 1100
 1011
 ----
  1111
  1011
  ----
   1000
   1011
   ----
     1100
     1011
     ----
      111   ← remainder
```

CRC remainder = 111.

(c) Complete transmitted bit string. (2)

110101111 (data 110101 followed by CRC 111). Check: 110101111 ÷ 1011 → remainder 000 ✔.

Q4. Simplified Internet checksum, 4-bit words. Message 1010 1100 0110 1001; one's-complement addition with end-around carry.

(a) Show the addition and calculate the checksum. (3)

Step	Addition	Raw result	End-around carry	Running sum
1	1010 + 1100	1 0110	0110 + 1	**0111**
2	0111 + 0110	1101	none	**1101**
3	1101 + 1001	1 0110	0110 + 1	**0111**

Sum = 0111. Checksum = one's complement = 1000. (Decimal check: 10 + 12 + 6 + 9 = 37; mod 15 → 7 = 0111 ✔.)

(b) 4-bit result expected when the receiver adds all message words and the checksum, assuming no error. (1)

0111 + 1000 = **1111** (all ones; complementing it gives 0000, meaning no error detected).

Q5. Define a single-bit error and a burst error. Explain how the length of a burst error is measured. (3)

- **Single-bit error**: only **one bit** of the data unit changes (0 → 1 or 1 → 0). (1)

- **Burst error**: **two or more bits** in the data unit change (not necessarily consecutive). (1)
- **Burst length**: measured from the **first corrupted bit to the last corrupted bit, inclusive**. Bits between them may or may not be corrupted. E.g. errors at bit positions 3 and 6 → burst length 6 − 3 + 1 = **4**. (1)

Q6. Even-parity scheme with one parity bit per block. Is it guaranteed to detect each of the following? Justify.

(a) A single-bit error. (2)

Yes, always. Even parity makes the number of 1s even. Flipping **one** bit changes the count by exactly one, making it **odd**, so the receiver's check fails and the error is detected.

(b) A burst error. (2)

No, not guaranteed. Parity detects only an **odd** number of bit errors. A burst that flips an **even** number of bits (2, 4, ...) leaves the count of 1s even, so the block **passes** undetected. E.g. 1011 0010 → 1**10**1 0010: two errors, still four 1s. A burst is detected only if it happens to flip an odd number of bits.

(c) Can a single parity bit identify and correct the erroneous bit? (1)

No. It only shows that **an odd number of bits** changed, not **which** bit. Correction needs more redundancy (e.g. Hamming code or 2-D parity) or retransmission.

Q7. Bluetooth profiles add complexity. Explain two ways in which profiles benefit application developers or users. (4)

1. **Interoperability (users)** (2): profiles define a *"standard of interoperability so that products from different vendors that claim to support a given usage model will work together"* [BT-B]. A user can pair any compliant headset, car kit or keyboard with any phone without compatibility worries.
2. **Ready-made, reusable implementation (developers)** (2): a profile specifies **exactly which protocols and features are mandatory, optional or not applicable** for a usage model, and builds on other profiles. E.g. the File Transfer Profile reuses OBEX, RFCOMM, L2CAP and the Generic Access Profile [BT-B]. Developers implement a known recipe (and only the subset they need), which saves effort and keeps devices small, cheap and low-power.

Q8. Compare the Bluetooth Classic central (master) with an access point in an 802.11 infrastructure WLAN. Explain one difference in medium-access coordination and one in their roles in connecting devices to other networks. (4)

- **Medium-access coordination (2)**: the **Bluetooth central schedules all access**. It defines the hop sequence and timing, and **polls** slaves, who may transmit **only in the slot after being addressed** (FH-TDD-TDMA), so there is **no contention or collision inside the piconet** [BT-B]. In **802.11**, stations **contend using CSMA/CA (DCF)**: sense, DIFS, random backoff, RTS/CTS, NAV. **Collisions are possible**, and the AP does not normally schedule access (polling via PCF is optional) [WL 32–40].
- **Connecting to other networks (2)**: the **802.11 AP is a bridge/portal**: it *"provides a link to other networks and acts as a relay point for all MAC frames"*, connecting the BSS to the wired distribution system/Internet [BT-S 64; FZ]. The **Bluetooth central only sets up the piconet and relays traffic within it** [BT-S 64]. Slaves can reach other piconets only via **scatternet bridge devices**, and a piconet is a short-range PAN, not an infrastructure gateway.

Q9. Dijkstra's shortest-path algorithm on the given network.

Link costs read from the figure: A–B 2, A–G 6, B–C 7, B–E 2, E–G 1, E–F 2, F–C 3, F–H 2, G–H 4, C–D 3, H–D 2.

(a) Shortest path from A to all nodes (step by step). (6)

Step	N'	B	C	D	E	F	G	H
0	A	**2,A**	∞	∞	∞	∞	6,A	∞
1	AB	—	9,B	∞	**4,B**	∞	6,A	∞
2	ABE	—	9,B	∞	—	6,E	**5,E**	∞
3	ABEG	—	9,B	∞	—	**6,E**	—	9,G
4	ABEGF	—	9,B	∞	—	—	—	**8,F**
5	ABEGFH	—	**9,B**	10,H	—	—	—	—
6	ABEGFHC	—	—	**10,H**	—	—	—	—
7	ABEGFHCD	—	—	—	—	—	—	—

Working:

- Step 0: A's neighbours are B 2 and G 6. Smallest = **B**.
- Step 1: via B (2): C = 2 + 7 = 9; E = 2 + 2 = 4. Smallest = **E (4)**.
- Step 2: via E (4): G = 4 + 1 = **5** (< 6, update); F = 4 + 2 = 6. Smallest = **G (5)**.
- Step 3: via G (5): H = 5 + 4 = 9. Smallest = **F (6)**.
- Step 4: via F (6): C = 6 + 3 = 9 (equal to 9, keep B); H = 6 + 2 = **8** (< 9, update). Smallest = **H (8)**.
- Step 5: via H (8): D = 8 + 2 = 10. Smallest = **C (9)**.
- Step 6: via C (9): D = 9 + 3 = 12 > 10 (no change). Smallest = **D (10)**. Done.

(b) Final shortest-path cost and complete route from A to B, C, D, E and F. (4)

Destination	Cost	Route
B	**2**	A → B
C	**9**	A → B → C *(equal-cost alternative A → B → E → F → C = 9)*
D	**10**	A → B → E → F → H → D
E	**4**	A → B → E
F	**6**	A → B → E → F

(Also: G = 5 via A → B → E → G; H = 8 via A → B → E → F → H.)

Test 1 — Version A (14 questions)

Q1. Explain the three basic propagation mechanisms in a mobile communication system. (12)

In a mobile system, the receiver often has **no clear line of sight** to the base station. The signal still arrives because of three mechanisms, which also cause **multipath** [WC-B 5.3, Fig 5.5]:

1. Reflection (4 marks): occurs when the EM wave meets a surface that is **large compared with its wavelength**, e.g. the ground, walls, buildings and furniture [WCP]. Part of the energy bounces off. In a mobile system the **ground-reflected wave** and the line-of-sight wave can **cancel each other**, causing high signal loss. Because the mobile antenna is lower than most buildings, many reflected waves arrive, and they combine **constructively or destructively** at the receiver. Reflection is the most significant effect even when LOS exists.

2. Diffraction (4 marks): occurs at the **edge of an impenetrable body that is large compared with the wavelength** (a sharp edge such as a building corner or rooftop). The edge acts as a **new source**: **secondary waves** spread out in different directions, including **into the shadow behind the obstacle**. Signals can therefore be received **even when there is no unobstructed LOS** from the transmitter [WC-B 5.3; WCP].

3. Scattering (4 marks): occurs when the obstacle is **on the order of the wavelength or smaller** and there are **many such objects**: lamp posts, traffic signs, foliage, rough surfaces. The incoming signal is **scattered into many weaker outgoing signals** in all directions. At cellular frequencies such objects are everywhere, so **scattering is hard to predict** [WC-B 5.3].

Overall effect: with a clear LOS, diffraction and scattering are minor and reflection matters most. With **no LOS** (e.g. at street level in a city), **diffraction and scattering are the primary means of reception**. Together the three produce **multipath**: several delayed copies that cause **fading and ISI** [WC-B 5.3].

Q2. What are the main problems of signal propagation? Why do radio waves not always follow a straight line? Why is reflection both useful and harmful? (12)

Main problems (6 marks) [WC-S 18, 39; WCP §2.3; WC-B 5.2–5.3]:

1. **Attenuation / path loss / free-space loss**: power falls with distance ($\propto d^2$) and is greater at higher frequencies.
2. **Noise**: thermal (cannot be eliminated), intermodulation, crosstalk, impulse noise (the main source of digital errors).
3. **Atmospheric absorption**: water vapour and oxygen attenuate the signal.
4. **Multipath propagation**: reflection, diffraction and scattering create delayed copies.
5. **Fading**: fast/slow and flat/frequency-selective variation in received power; shadowing.
6. **Intersymbol interference**: delayed copies overlap the next symbols.
7. **Refraction**: bending of the wave.
8. **Interference** from other transmitters (co-channel, adjacent-channel), plus the shared medium and a higher BER.

Why waves don't always travel in straight lines (3 marks):

- **Refraction**: wave speed depends on the **density of the medium**, so at the boundary between media (or through the varying atmosphere) the wave **bends**. Sky waves (3–30 MHz) are **refracted by the ionosphere** back to Earth [WC-S 15, 19].
- **Ground waves** (below ~3 MHz) **follow the curvature of the Earth** [WC-S 14].
- **Reflection, diffraction and scattering** redirect the wave around and off obstacles [WC-B 5.3].

Reflection: useful and harmful (3 marks):

- **Useful**: reflected waves let the signal **reach receivers with no line of sight** (indoors, behind buildings, in streets). The extra copies can be **combined by diversity/MIMO** receivers to strengthen the signal [WC-B 5.3–5.4].
- **Harmful**: reflected copies arrive with **different delays and phases**, so they can **cancel** the direct signal (the ground reflection vs LOS), causing **fading** (20–30 dB dips), **distortion** and **intersymbol interference**, which raises the bit error rate [WC-B 5.3].

Q3. Differences between path loss, multipath propagation and interference from other sources. (9)

Path loss is the **reduction in signal power** as the signal travels from transmitter to receiver. In free space it happens because energy **spreads over an ever-larger sphere**, so $Pt/Pr = (4\pi d/\lambda)^2$: loss grows with the **square of distance and of frequency**. Real environments add obstruction and absorption, described by the **path-loss exponent** n (2 in free space, 2.7–3.5 urban, 4–6 obstructed indoors). Path loss is **predictable and deterministic**. It sets the transmit power, receiver sensitivity and SNR needed, and it is countered by **higher power, better antennas or more access points** [WC-B 5.2; WC-S 30–34; WCP §9].

Multipath propagation happens when **reflection, diffraction and scattering** from buildings, walls, the ground and small objects make **several copies of the same signal arrive with different delays and phases**. The copies add **constructively or destructively**, causing **fading** (drops of 20–30 dB over half a wavelength) and **intersymbol interference**. It **varies randomly with time and position** as the user moves, and gives "high signal strength but poor quality". It is countered by **equalization, diversity/MIMO, OFDM and spread spectrum** [WC-B 5.3; WCP §7].

Interference from other sources is caused by **other transmitters**, not the wanted signal's path. Examples are **co-channel** interference (other WLANs on the same channel, microwave ovens at 2.4 GHz), **adjacent-channel** interference, harmonics and atmospheric noise. It **depends on other users' activity**, lowers the SINR, and causes **bit errors and blocked transmissions**. It is countered by **channel/frequency planning, spread spectrum, directional antennas and removing the source** [WCP §10; WC-S 10].

	Path loss	Multipath	Interference
Cause	Distance/frequency spreading	Reflection, diffraction, scattering of **own** signal	**Other** transmitters
Behaviour	Deterministic, smooth	Random, time-varying (fading, ISI)	Random/bursty, depends on others
Fix	Power, antennas, more APs	Equalization, diversity, OFDM	Channel planning, spread spectrum

Additional points for full marks:

- **Path loss**: example: indoors it increases ~**20 dB per 100 ft**; the main countermeasure is **additional access points** [WCP §9]. Expressed as $Lp = (4\pi d/\lambda)^2 = Pt/Pr$.
- **Multipath**: it is especially severe **indoors**; vendors use **equalization and antenna diversity** and may need **lower data rates** [WCP §7]. It produces **Doppler** effects when the receiver moves.
- **Interference**: two directions: **inward** (co-channel, from other systems on similar frequencies, e.g. microwave ovens at 2.4 GHz) and **outward** (adjacent-channel: your WLAN disrupting others, e.g. aircraft navigation). It is uncommon in spread-spectrum WLAN bands because power is low (< 1 W), so devices must be close and in the same band [WCP §10].

Q4. Explain the spread spectrum principles of (i) FHSS and (ii) DSSS. (10)

General principle (2 marks): spread spectrum is designed for wireless, where stations share the air and need protection from **eavesdropping and jamming**. It adds **redundancy** by expanding bandwidth **B to Bss ≫ B**. Two principles: (1) the allocated bandwidth is **much larger than needed**; (2) the spreading is done **after** the signal is created, by a process **independent of the signal**, using a **spreading code** that looks random but is a pattern [SS].

(i) FHSS (4 marks):

- Uses **M carrier frequencies**. At each moment the source modulates **one carrier**, and the next moment another.
- A **pseudorandom code generator** produces a **k-bit pattern each hopping period Th** ($M = 2^k$); a **frequency table** maps it to a frequency; a **frequency synthesizer** generates it; the **modulator** applies the data.
- *Example*: M = 8, k = 3; the pattern 101, 111, 001, ... selects 700 kHz, 900 kHz, 300 kHz, ... and repeats after 8 hops.
- **Privacy**: an intruder can only capture a **small piece** without the sequence. **Anti-jamming**: a jammer can hit **one hop, not the whole transmission**. **Sharing**: M stations can share Bss, like FDM with the allocation changing every hop [SS].

(ii) DSSS (4 marks):

- **Each data bit is replaced by n bits called chips**, using a spreading code. The **chip rate is n times the bit rate**, so the bandwidth grows n times.
- *Example*: the **Barker sequence** (n = 11, **10110111000**) used in wireless LANs. With polar NRZ, the spread signal = data × chips (a 1 sends the code, a 0 sends its inverse). The required bandwidth is **11×** the original.
- **Privacy** if the intruder doesn't know the code; **immunity to interference** if each station uses a different code. Bandwidth can be **shared** only if special (orthogonal) codes are used, i.e. **CDMA** [SS].

Q5. Benefits of the OFDMA approaches: adjacent, regularly spaced, randomly spaced subcarriers. (9)

In OFDMA, subcarriers are grouped into **subchannels** that are allocated to users. The three ways of forming subchannels each have benefits [OF §8.2]:

(i) Adjacent subcarriers (a contiguous block): all subcarriers in the block have **about the same SINR**. This creates an **opportunity**: the system can **choose, from many possible blocks, the best block for each user**, optimising **channel efficiency, user requirements, priority and fairness**. It exploits frequency-selective fading for multi-user gain. (The cost is that it needs accurate, up-to-date **channel knowledge across the whole bandwidth**, via pilots and prediction; used by LTE and WiMAX.)

(ii) Regularly spaced subcarriers ("distributed"/"comb"): spreading a user's subcarriers across the band gives **frequency diversity**, so **enough good subcarriers are obtained whichever are chosen**, even in frequency-selective fading. The **channel-estimation burden is much lower** (used by LTE).

(iii) Randomly spaced subcarriers: the **same frequency-diversity benefit** as regular spacing, **plus reduced adjacent-cell interference**, because neighbouring cells' random patterns rarely coincide (used by WiMAX).

Additional detail:

- **Adjacent**: all SINRs are about equal, so whole blocks can be compared. The scheduler can pick **each user's best block** (optimising efficiency, requirements, priority, fairness). It requires **pilot symbols, channel knowledge across the full band, and prediction**, because information may be out of date. Used by **LTE and WiMAX** [OF].
- **Regularly spaced**: "distributed"/"comb"; **frequency diversity**, so there are enough good subcarriers **regardless of which are chosen**; **low channel-estimation burden**. Used by **LTE**.
- **Randomly spaced**: the **diversity benefits** of regular spacing **plus reduced adjacent-cell interference**. Used by **WiMAX**.

Q6. Describe CDMA and explain why it is difficult to jam. (4)

CDMA lets **all users transmit over the entire bandwidth at the same time**. Each user is separated by a **unique pseudorandom (orthogonal) code** that **spreads** its baseband data at a much higher **chip rate**. The spread signal is transmitted **below the noise level**. The receiver uses a **correlator with the same code to despread** the wanted signal, which then passes a narrowband filter. Other users' signals are **not despread** and remain low-level noise. **Precise power control** keeps every transmitter at the minimum power [MA p7–8].

Difficult to jam because:

1. The energy is **spread over a very wide band at very low power density**, so a jammer must cover the whole band to have an effect.
2. **Despreading spreads the jammer**: the correlator collapses the wanted signal back to narrowband but **spreads the jamming signal** (it doesn't carry the user's code), and the **narrowband filter removes most of it**.
3. The **pseudorandom code is unknown** to the jammer, so it cannot concentrate its energy where the signal is.

That is why spread-spectrum CDMA was used in **WWII to foil enemy jamming** [MX], and why its stated uses include **suppressing interference and making interception difficult** [MA p7].

Q7. As a mobile node gets farther from a base station, what two actions could the BS take so that the frame loss probability does not increase? (4)

1. **Increase transmit power** (2 marks): received power falls as **$1/d^2$** ($Pr = Pt/(4\pi d/\lambda)^2$). Raising Pt keeps Pr and the SNR above the receiver's required threshold [WCP Eq 2.1–2.2].
2. **Lower the data rate / use more robust modulation and coding** (2 marks): with **adaptive modulation and coding** (802.11 **rate adaptation**), the BS switches to a lower-order modulation and stronger FEC, which need a **lower SNR for the same bit error rate** [WC-S 51; WL 23]. (Also acceptable: **beamforming** a directional beam toward the node [WC-B 5.4].)

Q8. Use Figure 2 to explain the hidden node problem. What happens when A and C send simultaneously? (4)

In Figure 2, **B is within range of both A and C, but A and C are outside each other's range**. When A transmits to B, **C cannot hear A**, so C's carrier sense finds the channel idle. A and C are **hidden from each other with respect to B** [FZ]. **If A and C send at the same time** (e.g. both to B): both sense an idle medium and transmit. Their signals **overlap at B, so B receives a collision** and can decode neither frame. Since wireless stations **cannot detect collisions**, A and C keep sending their **whole frames**, wasting the channel. Neither receives an **ACK**, so both time out, **back off a random time and retransmit**. Hidden nodes therefore **reduce network capacity** [FZ; WL 29].

Q9. Use Figure 2 to explain how CSMA/CA is realised in 802.11 (A sends). (4)

1. **Carrier sense**: A checks its **NAV** and senses the medium. When idle, it waits **DIFS**, then counts down a **random backoff** (the counter freezes if the medium becomes busy) [WL 32–35].
2. **RTS**: A sends an RTS containing the **duration** of the whole exchange. B hears it (C doesn't).
3. **CTS**: after **SIFS**, B replies with a CTS that also carries the duration. **C is in B's range, so it hears the CTS, sets its NAV and defers**, even though it can't hear A. This is how the hidden node is handled [FZ; WL 39, 43].
4. **DATA and ACK**: after SIFS, A sends the data. After SIFS, B returns an **ACK**. If A gets no CTS or no ACK, it assumes a collision, backs off and retries [WL 35, 38].

Q10. Why are acknowledgements used in 802.11 but not in wired Ethernet? (4)

1. **No collision detection in wireless**: a station can't transmit and receive at the same time (the received signal is far weaker; it would need costly full-duplex hardware) [WL 16, 29].
2. **Hidden stations**: even if it could listen, a collision at the receiver may be **inaudible to the sender** [FZ].
3. **High error rate**: noise, interference, fading and multipath corrupt many frames, so the MAC and PHY are **unreliable** [WL 27].
4. **In Ethernet**, a **detected** collision triggers a resend and **no collision acts as an implicit acknowledgement**, on a low-error cable [FZ]. So 802.11 uses **explicit ACKs after SIFS**, and retransmits when no ACK arrives [WL 27, 34].

Q11. Two ISPs in a café. (i) Both on channel 11. (4) (ii) Channels 1 and 11. (4)

(i) By accident both APs use channel 11. Will 802.11 completely break down? What happens when two stations, each associated with a different ISP, transmit at the same time? (4)

No, it does not break down. The two BSSs simply **share one channel as a single collision domain**. Every station still runs **CSMA/CA**: it senses the channel and its NAV, so stations from **both** ISPs **defer to each other** and take turns, as if they were one bigger BSS. (The different IP blocks don't matter at the MAC layer; each AP only accepts frames addressed to its own BSS.) If two stations from different ISPs **transmit at the same moment** (e.g. their backoff counters expire together, or they are hidden from each other), their frames **collide**. Neither AP receives a correct frame, **no ACKs are returned**, both stations **back off a random time** (with a larger contention window) and **retransmit**. The result is **more collisions and lower throughput for both ISPs** (each gets roughly half the channel capacity), but the network keeps working.

(ii) Now one AP uses channel 1 and the other channel 11. How do the answers change? (4)

Channels **1 and 11 do not overlap** (in the 2.4 GHz band, signals are ~22 MHz wide [WC-S 4] and channels are 5 MHz apart, so 1 and 11 are 50 MHz apart; SUPPLEMENTARY). The two BSSs now form **two independent collision domains**:

- Stations of different ISPs **don't hear each other's transmissions** in carrier sense, so they **don't defer to each other**.
- **Two stations can transmit at the same time without colliding**: each frame reaches its own AP successfully and is ACKed.
- **Each ISP gets the full capacity of its own channel**: aggregate throughput roughly **doubles**, with fewer collisions and backoffs.

Q12. A wireless receiving site is experiencing occasional loss of signal due to interference. Discuss the steps you would take to correct this problem. (10)

A systematic approach (about 1 mark per explained step) [WCP §9–10; WC-S 9–10; WL 12, 21–23; WC-B 5.4]:

1. **Characterise the problem**: record **when** it happens (time of day, which events), **where** (which areas), and **which channels/frequencies**. "Occasional" loss suggests an **intermittent source**, e.g. a **microwave oven at 2.4 GHz** or a neighbouring network's bursts.
2. **Survey the spectrum**: use a **spectrum analyser / WLAN site-survey tool** to identify interfering signals and **co-channel or adjacent-channel** networks. Decide whether it is **inward** interference (others → us) or something we cause (outward).
3. **Identify and locate the source**: microwave ovens, cordless phones, Bluetooth devices, neighbouring APs, other ISM equipment, faulty electrical equipment (impulse noise).
4. **Remove, switch off, shield or relocate** the interfering device, or **increase the separation** between it and the receiver. Interference needs the devices to be **close and in the same band** [WCP §10].
5. **Change channel**: move the AP/link to a **clear, non-overlapping channel** (e.g. 1, 6 or 11 in 2.4 GHz), or move to the **less crowded 5 GHz band** [WL 23].
6. **Improve the antenna system**: reposition/raise the antenna to restore LOS; use **directional antennas** that point at the wanted transmitter and **reject** interference from other directions; use **antenna/space diversity** [WC-B 5.4].
7. **Improve the link margin**: increase the **wanted signal's power or antenna gain** within regulatory limits, so the **SINR** stays above threshold during interference bursts.
8. **Use robust transmission techniques**: **spread spectrum / (adaptive) frequency hopping**, **FEC**, **adaptive modulation and coding / lower data rates**, and ARQ to ride out bursts [WC-B 5.4].
9. **Network design**: add APs or reduce cell size; coordinate channel and power plans with neighbouring networks.
10. **Verify and monitor**: re-test after each change, and keep monitoring to confirm the problem has gone.

Q13. Explain why many WiFi networks can have a high error rate caused by co-channel interference. (4)

1. **Few clean channels**: 2.4 GHz has only **three non-overlapping channels** (1, 6, 11; SUPPLEMENTARY), and the **AP administrator chooses the channel**, so *"interference [is] possible: channel can be same as that chosen by neighboring AP"* [WL 12]. Dense deployments put many BSSs on the same channel.
2. **Crowded, unprotected ISM band**: 2.4 GHz is shared with cordless phones, Bluetooth and **microwave ovens**. ISM users **must tolerate interference** and have **no regulatory protection** [WL 21, 23; WC-S 10].
3. **Same-frequency signals add directly to the wanted signal**: co-channel (inward) interference **blocks transmissions or causes bit errors** [WCP §10], lowering the SINR.
4. **Hidden stations in neighbouring BSSs** on the same channel cause collisions that carrier sensing can't prevent. The result is lost frames, missing ACKs, retransmissions and lower throughput.

Q14. Transmitter at 850 MHz, power 125 mW; received power 1 µW. Distance between transmitter and receiver? (6)

Given: Pt = 125 mW = 125×10^{-3} W; Pr = 1 µW = 1×10^{-6} W; f = 850×10^{6} Hz; c = 3×10^{8} m/s.

Formula: $L_p = P_t/P_r = (4\pi d/\lambda)^2$ [WCP Eq 2.1, 2.3].

1. **Path loss**: $L_p = P_t/P_r = (125 \times 10^{-3})/(1 \times 10^{-6}) =$ **$125 \times 10^3 = 125\,000$**
2. **Wavelength**: $\lambda = c/f = (3 \times 10^8)/(850 \times 10^6) =$ **0.3529 m**
3. **Substitute**: $125\,000 = (4\pi d/0.3529)^2$
4. **Square root**: $\sqrt{125\,000} =$ **353.55** $= 4\pi d/0.3529$
5. **Solve**: $d = (353.55 \times 0.3529)/(4\pi) = 124.77/12.566 =$ **9.93 m**

Answer: $d \approx 9.93$ m (this is Problem 2.7 in your notes. Note that the notes' printed step "125×10^{-3}" should read 125×10^3).

COM5231 Test 1 — Version B (100 marks)

Q1. Explain the difference between single-bit and burst errors and the extent to which parity checking is appropriate as an error-detecting technique in both cases. (10)

Definitions SUPPLEMENTARY (standard; the worksheet [ED] lists this question but gives no theory). Causes from [WC-B].

Single-bit error (2 marks): **only one bit** of a data unit (byte, frame) is changed, from 0 to 1 or 1 to 0. E.g. 0000 0010 sent, 0000 1010 received. These are relatively **rare** in practice, because noise usually lasts longer than one bit time.

Burst error (3 marks): **two or more bits** in the data unit are changed. They need not be consecutive. The **burst length** runs from the **first** corrupted bit to the **last**, inclusive. Bursts are the **common case** in wireless: **impulse noise** (*"the primary source of error in digital data... a 0.01 s spike... would wash out about 10 000 bits at 1 Mbps"* [WC-B 5.2]) and **fading** dips of 20–30 dB [WC-B 5.3] corrupt many consecutive bits.

Parity checking (even parity: one bit appended so the number of 1s is even; the receiver recounts):

- **Single-bit errors (2 marks)**: parity **always detects** them. One flipped bit changes the count of 1s from even to odd. It is **cheap** (one extra bit) and simple, so it is **appropriate** where errors are isolated. It **cannot correct** them, though: it can't tell **which** bit flipped, so retransmission is needed.
- **Burst errors (3 marks)**: parity detects a burst **only if an odd number of bits** is flipped. Any **even** number of errors (2, 4, ...) leaves the parity unchanged and goes **undetected**. E.g. 10110010 → 1**10**10010: two errors, still four 1s, so it passes. Roughly **half of all bursts escape**, so single parity is **not appropriate** for bursty wireless channels. **Two-dimensional parity** improves this (it detects many bursts and corrects single errors), but **CRC** is preferred, since it detects **all bursts of length ≤ r** (the number of check bits).

Q2. Transmit the message 10011010 protected by the CRC polynomial $x^3 + x^2 + 1$.

(a) Use polynomial long division to determine the message that should be transmitted. (10)

1. **Generator**: $x^3 + x^2 + 0{\cdot}x + 1$ → **1101**; degree r = **3**.
2. **Append 3 zeros**: $M(x){\cdot}x^3$ → **10011010 000**.
3. **Divide by 1101 (modulo-2)**:

Step	Leading window	XOR 1101	Result	Remaining dividend
1	1001	1101	0100	0100 1010 000
2	1001	1101	0100	0010 0010 000
3	1000	1101	0101	0001 0110 000
4	1011	1101	0110	0000 1100 000
5	1100	1101	0001	0000 0001 000
6	1000	1101	0101	0000 0000 101

```
1 0 0 1 1 0 1 0 0 0 0
1 1 0 1
-------
  1 0 0 1
  1 1 0 1
  -------
    1 0 0 0
    1 1 0 1
    -------
      1 0 1 1
      1 1 0 1
      -------
        1 1 0 0
        1 1 0 1
        -------
              1 0 0 0
              1 1 0 1
              -------
                1 0 1   = remainder
```

4. **Remainder (CRC) = 101.** Polynomial check: $x^{10} + x^7 + x^6 + x^4 \bmod (x^3 + x^2 + 1) = \mathbf{x^2 + 1} \leftrightarrow 101$ ✔.
5. **Transmitted message = 10011010 101**. Check: 10011010101 ÷ 1101 → remainder **000** ✔.

(b) The leftmost bit is inverted by noise. What is the result of the receiver's CRC calculation? How does the receiver know an error occurred? (6)

- Received: **0** 0011010101 = **00011010101**.
- The receiver divides by 1101. The leading zeros need no XOR; the first 1 is at position 4: 1101 ⊕ 1101 = 0000 → **00000000101**.
- **Remainder = 101 (≠ 000).**
- A correctly received frame is **exactly divisible by G** (remainder 0), because the sender made it so. A **non-zero remainder** therefore proves the frame was corrupted. The receiver **detects the error**, discards the frame and requests retransmission (ARQ). (This matches the worksheet's own example: *"Since the remainder is non-zero, we know that something went wrong!"* [ED].)

Q3. Explain the three basic propagation mechanisms in a mobile communication system. (12)

In a mobile system, the receiver often has **no clear line of sight** to the base station. The signal still arrives because of three mechanisms, which also cause **multipath** [WC-B 5.3, Fig 5.5]:

1. Reflection (4 marks): occurs when the EM wave meets a surface that is **large compared with its wavelength**, e.g. the ground, walls, buildings and furniture [WCP]. Part of the energy bounces off. In a mobile system the **ground-reflected wave** and the line-of-sight wave can **cancel each other**, causing high signal loss. Because the mobile antenna is lower than most buildings, many reflected waves arrive, and they combine **constructively or destructively** at the receiver. Reflection is the most significant effect even when LOS exists.

2. Diffraction (4 marks): occurs at the **edge of an impenetrable body that is large compared with the wavelength** (a sharp edge such as a building corner or rooftop). The edge acts as a **new source**: **secondary waves** spread out in different directions, including **into the shadow behind the obstacle**. Signals can therefore be received **even when there is no unobstructed LOS** from the transmitter [WC-B 5.3; WCP].

3. Scattering (4 marks): occurs when the obstacle is **on the order of the wavelength or smaller** and there are **many such objects**: lamp posts, traffic signs, foliage, rough surfaces. The incoming signal is **scattered into many weaker outgoing signals** in all directions. At cellular frequencies such objects are everywhere, so **scattering is hard to predict** [WC-B 5.3].

Overall effect: with a clear LOS, diffraction and scattering are minor and reflection matters most. With **no LOS** (e.g. at street level in a city), **diffraction and scattering are the primary means of reception**. Together the three produce **multipath**: several delayed copies that cause **fading and ISI** [WC-B 5.3].

Q4. Differences between path loss, multipath propagation and interference. (12)

Path loss is the **reduction in signal power** as the signal travels from transmitter to receiver. In free space it happens because energy **spreads over an ever-larger sphere**, so $Pt/Pr = (4\pi d/\lambda)^2$: loss grows with the **square of distance and of frequency**. Real environments add obstruction and absorption, described by the **path-loss exponent** n (2 in free space, 2.7–3.5 urban, 4–6 obstructed indoors). Path loss is **predictable and deterministic**. It sets the transmit power, receiver sensitivity and SNR needed, and it is countered by **higher power, better antennas or more access points** [WC-B 5.2; WC-S 30–34; WCP §9].

Multipath propagation happens when **reflection, diffraction and scattering** from buildings, walls, the ground and small objects make **several copies of the same signal arrive with different delays and phases**. The copies add **constructively or destructively**, causing **fading** (drops of 20–30 dB over half a wavelength) and **intersymbol interference**. It **varies randomly with time and position** as the user moves, and gives "high signal strength but poor quality". It is countered by **equalization, diversity/MIMO, OFDM and spread spectrum** [WC-B 5.3; WCP §7].

Interference from other sources is caused by **other transmitters**, not the wanted signal's path. Examples are **co-channel** interference (other WLANs on the same channel, microwave ovens at 2.4 GHz), **adjacent-channel** interference, harmonics and atmospheric noise. It **depends on other users' activity**, lowers the SINR, and causes **bit errors and blocked transmissions**. It is countered by **channel/frequency planning, spread spectrum, directional antennas and removing the source** [WCP §10; WC-S 10].

	Path loss	Multipath	Interference
Cause	Distance/frequency spreading	Reflection, diffraction, scattering of **own** signal	**Other** transmitters
Behaviour	Deterministic, smooth	Random, time-varying (fading, ISI)	Random/bursty, depends on others
Fix	Power, antennas, more APs	Equalization, diversity, OFDM	Channel planning, spread spectrum

Additional points for full marks:

- **Path loss**: example: indoors it increases ~**20 dB per 100 ft**; the main countermeasure is **additional access points** [WCP §9]. Expressed as $Lp = (4\pi d/\lambda)^2 = Pt/Pr$.
- **Multipath**: it is especially severe **indoors**; vendors use **equalization and antenna diversity** and may need **lower data rates** [WCP §7]. It produces **Doppler** effects when the receiver moves.
- **Interference**: two directions: **inward** (co-channel, from other systems on similar frequencies, e.g. microwave ovens at 2.4 GHz) and **outward** (adjacent-channel: your WLAN disrupting others, e.g. aircraft navigation). It is uncommon in spread-spectrum WLAN bands because power is low (< 1 W), so devices must be close and in the same band [WCP §10].

Q5. Discuss three methods to increase the capacity of a wireless communication system. (12)

Capacity = how many users or bits/s a system can carry in its spectrum. Three methods (4 marks each):

1. Frequency reuse with smaller cells (network densification / SDMA)

- Limit each transmitter's **range** and **reuse the same frequencies** in cells far enough apart that interference is acceptable. Traditional **macro cells** (km wide) are complemented by **small cells**: indoor **femtocells**, outdoor **picocells/relays**. Every reuse multiplies capacity, hence "network densification" [WC-B 5.4; WC-S 52].
- **SDMA/sectorization** strengthens this: directional antennas reduce co-channel interference, so the reuse pattern can be **tightened**. Moving from a **7-cell to a 4-cell pattern nearly doubles capacity**, and **intracell reuse** of the same channel by spatial separation becomes possible [MA p9–10].

2. Multiple-antenna techniques (MIMO)

- **SU-MIMO multilayer transmission** sends **parallel data streams** between one Tx–Rx pair, for a **higher data rate in the same bandwidth**.
- **MU-MIMO** forms **directional beams to several users simultaneously** (spatial multiple access), giving **more users per cell**.
- Diversity and beamforming improve link quality and coverage. 4×4 and 8×8 MIMO are in use; massive MIMO is coming [WC-B 5.4, Fig 5.9].

3. Bandwidth expansion

- **Carrier aggregation**: combine channels (802.11n/ac: 20 → 40/80/160 MHz; LTE aggregates carriers).
- **Millimetre-wave bands (30–300 GHz)**: much wider channels are available (at the cost of obstruction and absorption problems). These are needed because Shannon limits the bps/Hz achievable for a given SNR [WC-B 5.4; WC-S 52].

*(Other valid choices: **efficient multiple access**: CDMA gives 4–5× the capacity of other digital systems [MA p8] and OFDMA allows tight subcarrier packing and opportunistic scheduling [OF]; **adaptive modulation and coding** [WC-S 51].)*

Q6. Explain the spread spectrum principles of FHSS and DSSS. (10)

General principle (2 marks): spread spectrum is designed for wireless, where stations share the air and need protection from **eavesdropping and jamming**. It adds **redundancy** by expanding bandwidth **B to Bss ≫ B**. Two principles: (1) the allocated bandwidth is **much larger than needed**; (2) the spreading is done **after** the signal is created, by a process **independent of the signal**, using a **spreading code** that looks random but is a pattern [SS].

(i) FHSS (4 marks):

- Uses **M carrier frequencies**. At each moment the source modulates **one carrier**, and the next moment another.
- A **pseudorandom code generator** produces a **k-bit pattern each hopping period Th** ($M = 2^k$); a **frequency table** maps it to a frequency; a **frequency synthesizer** generates it; the **modulator** applies the data.
- *Example*: M = 8, k = 3; the pattern 101, 111, 001, ... selects 700 kHz, 900 kHz, 300 kHz, ... and repeats after 8 hops.
- **Privacy**: an intruder can only capture a **small piece** without the sequence. **Anti-jamming**: a jammer can hit **one hop, not the whole transmission**. **Sharing**: M stations can share Bss, like FDM with the allocation changing every hop [SS].

(ii) DSSS (4 marks):

- **Each data bit is replaced by n bits called chips**, using a spreading code. The **chip rate is n times the bit rate**, so the bandwidth grows n times.

- *Example*: the **Barker sequence** (n = 11, **10110111000**) used in wireless LANs. With polar NRZ, the spread signal = data × chips (a 1 sends the code, a 0 sends its inverse). The required bandwidth is **11×** the original.
- **Privacy** if the intruder doesn't know the code; **immunity to interference** if each station uses a different code. Bandwidth can be **shared** only if special (orthogonal) codes are used, i.e. **CDMA** [SS].

Q7. Describe CDMA and explain why it is difficult to jam. (10)

Description (6 marks):

1. **Spread-spectrum multiple access**: all users **share the entire bandwidth all the time**; there are no separate frequencies or time slots [MA p7; MX].
2. **Unique pseudorandom (orthogonal) code per user**, shared by the mobile and the base station [MA p8].
3. **Spreading**: data are XORed/multiplied with the code at a much higher **chip rate**. E.g. IS-95: **1.2288 Mchip/s** spreading 13 kbps voice over **1.25 MHz**, with up to **64 codes/users** per channel [AT].
4. **Despreading**: the receiver's **correlator** uses the same code. Only the wanted signal is despread and passes the **narrowband filter**; other users' signals remain noise-like [MA p8].
5. **Power control**: all users transmit at the minimum required power, to limit mutual interference (the near-far problem) and save battery [MA p8].
6. **Advantages**: greatest spectrum efficiency (8–10× analogue, 4–5× other digital), **soft handoff**, simplified frequency planning, privacy (Walsh codes). Used in IS-95 and 3G [MA p8].

Why it is difficult to jam (4 marks):

1. The signal occupies a **very wide band at a power density below the noise floor**. A jammer would need enormous power across the whole band.
2. **Processing gain**: despreading **spreads the jammer** over the wide band while concentrating the wanted signal. The narrowband filter then rejects most of the jamming energy (SUPPLEMENTARY: gain ≈ chip rate/bit rate, e.g. ≈ 21 dB for IS-95).
3. The jammer **doesn't know the pseudorandom code** (or hop sequence), so it cannot correlate with or mimic the signal. In FH, *"a malicious sender may jam one hopping period, but not the whole period"* [SS].
4. **Designed for anti-jamming**: spread spectrum's purposes include suppressing interference and making interception difficult [MA p7]. CDMA was a **military technology used in WWII to foil German jamming** [MX].

Q8. Describe the difference between OFDM and OFDMA. (10)

OFDM (4 marks): a **multicarrier modulation/transmission technique**. A high-rate stream (R bps) is split by a serial-to-parallel converter into **N lower-rate substreams (R/N)**, carried on **N orthogonal subcarriers** (spacing fb = a multiple of 1/T, e.g. 15 kHz in LTE). Orthogonality lets the spectra **overlap without interference** (≈ 6× more signals than FDM in the notes' example). Longer symbols (N·Ts) **defeat ISI** and **frequency-selective fading** hits only some subcarriers. **In OFDM all subcarriers carry the data of a single source** [OF §8.1].

OFDMA (4 marks): a **multiple access scheme built on OFDM**: a combination of **FDMA and TDMA**, where **different users are assigned subsets of the subcarriers (subchannels / LTE resource blocks of 12 subcarriers) at different times**. Subchannels can be formed from **adjacent, regularly spaced or randomly spaced** subcarriers, and are allocated by **opportunistic scheduling** (efficiency, proportional

fairness $r_i/\bar{r}_i$, requirements, priority). In the uplink, a user **concentrates power** on its own subchannels, saving battery [OF §8.2; AT].

Key differences (2 marks):

	OFDM	OFDMA
Type	Modulation / transmission technique	Multiple access technique
Users per symbol	One user uses all subcarriers	Many users share the subcarriers simultaneously
Multiple access	Needs another method (e.g. **CSMA** in 802.11, TDMA)	Built in (frequency + time)
Examples	802.11a/g/n	**LTE downlink**, WiMAX

"Not all technologies that use OFDM use OFDMA. Some forms of 802.11 use OFDM for the signal but CSMA for multiple access" [OF §8.2].

Q9. What distinguishes single-user MIMO from multiple-user MIMO? (8)

MIMO uses multiple antennas at transmitter and receiver, for diversity, beamforming, multi-user MIMO and multilayer transmission [WC-B 5.4].

SU-MIMO (4 marks): all spatial streams go between the base station/AP and **one user** at a time. **Multilayer transmission** sends **multiple parallel data streams** over the same frequency and time between that single Tx–Rx pair. **Goal: a higher data rate for that user in a given bandwidth.** The user device needs **multiple antennas** (number of streams ≤ min(Tx, Rx antennas)).

MU-MIMO (4 marks): *"with enough MIMO antennas, directional antenna beams can be established to **multiple users simultaneously**"* [WC-B 5.4]. The BS **beamforms** separate streams to **different users** on the same time/frequency: **spatial division multiple access**. **Goal: higher system capacity, i.e. more users per cell.** Users can have single antennas, but the BS needs **many antennas** and **knowledge of each user's channel** to steer the beams and cancel inter-user interference.

Summary: SU-MIMO increases **one user's throughput**; MU-MIMO increases **the number of users served simultaneously**, i.e. cell capacity [WC-B Fig 5.9].

Test 2 2025 (100 marks)

Q1. State and explain the three categories of routing protocols in MANETs. (12)

SUPPLEMENTARY (no lecture notes provided), plus [WL 8–9] and [BT-S 87].

Why it is complicated (4 marks):

1. **Dynamic topology**: nodes are **mobile**, so links break and form continuously and routes go stale quickly.
2. **No infrastructure or central controller**: there are no dedicated routers or APs. **Every node must also act as a router** and forward others' packets [WL 8–9], and decisions are distributed.
3. **Limited, shared wireless bandwidth**: routing control traffic competes with data, so flooding updates is expensive.
4. **Limited battery and processing power** in mobile nodes.
5. **Unreliable links**: fading, interference, **hidden/exposed terminals**, high error rates and asymmetric links; plus multi-hop delay and security concerns.

Three categories (6 marks):

1. **Proactive (table-driven)**: every node **maintains routes to all destinations at all times** using **periodic/triggered updates**. Routes are **available immediately** (low latency), but **control overhead is high and constant**, and wasted on routes never used and under high mobility. *Example: **DSDV** (Destination-Sequenced Distance Vector); also OLSR.*
2. **Reactive (on-demand / source-initiated)**: routes are found **only when a source needs one**, through **route discovery** (flood a route request; the destination replies) and **route maintenance**. **Low overhead** when traffic is light, but a **discovery delay** before the first packet. *Example: **AODV** (Ad hoc On-demand Distance Vector), which ZigBee uses: it "sends out route requests and either responds or forwards the messages to other nodes until someone responds" [BT-S 87]; also DSR.*
3. **Hybrid**: **proactive within a local zone** around each node and **reactive between zones**, which balances latency and overhead and scales to large networks. *Example: **ZRP** (Zone Routing Protocol).*

Additional detail for 12 marks:

- **Proactive**: works like **distance-vector/link-state routing adapted to mobility**. DSDV adds **destination sequence numbers** to avoid loops and count-to-infinity, with full and incremental table dumps. **OLSR** uses multipoint relays to reduce flooding. **Best for** small networks with low mobility and frequent traffic.
- **Reactive**: **AODV**: RREQ broadcast → reverse path set up → RREP unicast back → forward route installed; **RERR** on link breaks; sequence numbers keep routes fresh [BT-S 87 for the outline]. **DSR**: the **whole source route** is carried in each packet; route caching. **Best for** large, highly mobile networks with occasional traffic.
- **Hybrid**: **ZRP**: each node has a **zone** (radius in hops). Intra-zone routing is proactive (IARP); inter-zone routing is reactive (IERP), querying only **border nodes** ("bordercasting"). Combines low latency nearby with low overhead far away.

Q2. Dijkstra on Figure 1 from x to all nodes. (8)

Step	N'	D(t),p	D(u),p	D(v),p	D(w),p	D(y),p	D(z),p
0	x	∞	∞	**3,x**	6,x	6,x	8,x
1	xv	7,v	**6,v**	—	6,x	6,x	8,x

Step	N'	D(t),p	D(u),p	D(v),p	D(w),p	D(y),p	D(z),p
2	xvu	7,v	—	—	**6,x**	6,x	8,x
3	xvuw	7,v	—	—	—	**6,x**	8,x
4	xvuwy	**7,v**	—	—	—	—	8,x
5	xvuwyt	—	—	—	—	—	**8,x**
6	xvuwytz	—	—	—	—	—	—

Updates: via v (3): t = 3 + 4 = 7; u = 3 + 3 = 6; w via v = 7 > 6 and y via v = 11 > 6 (no change). Ties at 6 (u, w, y) broken alphabetically. Via u: t via u = 8 > 7; w via u = 9 > 6. Via y: t = 13, z = 18 (no change).

To	v	u	w	y	t	z
Cost	3	6	6	6	7	8
Path	x-v	x-v-u	x-w	x-y	x-v-t	x-z
First hop	v	v	w	y	v	z

Q3. What is the count-to-infinity problem in distance-vector routing? (5)

Count to infinity: in distance-vector routing, **bad news travels slowly**. When a link cost increases or a link fails, a node may switch to a neighbour's **stale** advertised route, which in fact passes **back through itself**. This creates a **routing loop**: the two nodes repeatedly re-advertise, each raising its estimate by a small amount per exchange, "**counting up**" to the true cost, or **to infinity** if the destination is unreachable. *Example*: y–x = 4, y–z = 1, z–x = 50. If y–x rises to 60, y picks the path via z (1 + 5 = 6), z picks via y (7), and so on for 44 exchanges, with packets looping between y and z. (Fixes: poisoned reverse, split horizon, a maximum hop count.)

- **Cause**: DV nodes know only their **neighbours' distance estimates**, not the paths behind them. After a link failure or cost increase, a node can adopt a neighbour's route that **goes back through itself**.
- **Effect**: a **routing loop**. Estimates rise by small steps with each exchange (**"bad news travels slowly"**), up to the true cost or to **infinity** if unreachable. Packets loop meanwhile. In the y/z/x example (4, 1, 50 → 60), it takes **44 iterations**.
- **Fixes**: **poisoned reverse** (advertise ∞ to the neighbour you route through), **split horizon**, a **maximum metric** (RIP: 16 = infinity).

Q4. Explain the hidden terminal problem. Does it occur in Bluetooth? (10)

Hidden terminal problem (6 marks): in a wireless LAN, station **B** is in range of both **A** and **C**, but **A and C cannot hear each other** (distance or an obstacle) [FZ Fig 15.3].

- A sends to B. C senses the medium, **hears nothing** and concludes it is idle, so it also transmits (to B). The two frames **collide at B**.
- Neither A nor C can **detect** the collision: wireless stations can't send and listen at once, and they can't hear each other anyway. So the whole frames are wasted, followed by ACK timeouts and retransmissions. **Capacity drops** [FZ].
- This is one of the three reasons **CSMA/CD can't be used** in wireless [WL 29].
- **Solution in 802.11**: the **RTS/CTS handshake**. The **CTS from B** (with the duration) reaches C, which sets its **NAV** and defers [FZ; WL 39, 43].

In Bluetooth (4 marks): **No, not within a piconet.** Access is **centrally scheduled by the master** (FH-TDD-TDMA polling). A slave transmits **only in the slot immediately after the master addresses it**, and

SCO slots are **reserved** [BT-B]. Since no device transmits on its own after sensing the channel, two hidden slaves can never start transmitting at the same time to the master. (Collisions between **different piconets** happen only when they hop to the same frequency in the same slot. That is a separate FH-CDMA effect handled by FEC/ARQ [BT-B].)

Q5. Compare wired and wireless LANs. What unique concerns must the designer of a wireless LAN address? (10)

Comparison (5 marks):

Aspect	Wired LAN	Wireless LAN
Medium	Wires; dedicated/switched links	**Air**; the signal is **broadcast** and **shared** by all hosts
Hosts	Fixed attachment point	**Not physically connected**; can **move freely**
Isolated LAN	Hosts joined by a link-layer **switch**	**Ad hoc network**: hosts communicate directly, **no AP**
Connection to other networks	Via a **router**	Via an **access point** that joins the wireless and wired environments
Moving between them	Change the NICs and replace the switch with an AP: **MAC addresses change, IP addresses stay the same**	
Access method	**CSMA/CD**	**CSMA/CA**
Channel	Negligible attenuation and interference	**Attenuation, interference, multipath, higher error rate**

Also from [WL 3]: wireless gives flexibility, ad hoc operation and disaster robustness, but lower bandwidth, lower QoS and higher delay.

Unique concerns (5 marks; 1 mark each, any five):

1. **Attenuation/path loss and coverage**: AP placement; more APs [WCP].
2. **Interference** in the crowded, unprotected **ISM band**: channel planning, 5 GHz [WL 23].
3. **Multipath and fading**: diversity, MIMO, OFDM, rate adaptation [WCP; WL 26].
4. **Higher error rates**: ACKs, retransmission, FEC, **fragmentation** [FZ; WL 27].
5. **Medium access without collision detection**: **hidden and exposed stations**, so CSMA/CA, RTS/CTS and NAV [FZ].
6. **Mobility/roaming** between BSSs and ESSs; association [FZ; WL 12].
7. **Power management** for battery devices [WL 4].
8. **Security and privacy** on a broadcast medium [WL 4, 27].
9. **Regulation**: licence-free but power-limited (≤ 1 W); national restrictions [WL 3, 23].
10. **QoS** for time-sensitive traffic: PCF, rate adaptation [WL 40].

Q6. A wireless receiving site has occasional loss of signal due to interference. Discuss the steps you would take. (10)

A systematic approach (about 1 mark per explained step) [WCP §9–10; WC-S 9–10; WL 12, 21–23; WC-B 5.4]:

1. **Characterise the problem**: record **when** it happens (time of day, which events), **where** (which areas), and **which channels/frequencies**. "Occasional" loss suggests an **intermittent source**, e.g. a

microwave oven at 2.4 GHz or a neighbouring network's bursts.

2. **Survey the spectrum**: use a **spectrum analyser / WLAN site-survey tool** to identify interfering signals and **co-channel or adjacent-channel** networks. Decide whether it is **inward** interference (others → us) or something we cause (outward).
3. **Identify and locate the source**: microwave ovens, cordless phones, Bluetooth devices, neighbouring APs, other ISM equipment, faulty electrical equipment (impulse noise).
4. **Remove, switch off, shield or relocate** the interfering device, or **increase the separation** between it and the receiver. Interference needs the devices to be **close and in the same band** [WCP §10].
5. **Change channel**: move the AP/link to a **clear, non-overlapping channel** (e.g. 1, 6 or 11 in 2.4 GHz), or move to the **less crowded 5 GHz band** [WL 23].
6. **Improve the antenna system**: reposition/raise the antenna to restore LOS; use **directional antennas** that point at the wanted transmitter and **reject** interference from other directions; use **antenna/space diversity** [WC-B 5.4].
7. **Improve the link margin**: increase the **wanted signal's power or antenna gain** within regulatory limits, so the **SINR** stays above threshold during interference bursts.
8. **Use robust transmission techniques**: **spread spectrum / (adaptive) frequency hopping**, **FEC**, **adaptive modulation and coding / lower data rates**, and ARQ to ride out bursts [WC-B 5.4].
9. **Network design**: add APs or reduce cell size; coordinate channel and power plans with neighbouring networks.
10. **Verify and monitor**: re-test after each change, and keep monitoring to confirm the problem has gone.

Q7. While Bluetooth's profiles add significant complexity, how can they benefit applications from the application's perspective? (10)

1. **Guaranteed interoperability between vendors**: the profile specification exists *"to define a standard of interoperability so that products from different vendors that claim to support a given usage model will work together"* [BT-B]. A user's headset, car kit or keyboard works with any phone or PC supporting the same profile.
2. **A complete recipe for each usage model**: a profile is *"a set of protocols that implement a particular Bluetooth-based application"*. It states which parts of the core specification are **mandatory, optional and not applicable** [BT-B]. Application developers don't have to interpret the 2500+ page core themselves; the design decisions are made for them.
3. **Reuse of existing protocols and other profiles**: profiles build on each other. E.g. the **File Transfer Profile depends on the Generic Object Exchange, Serial Port and Generic Access profiles**, using **OBEX** over **RFCOMM** and **L2CAP** [BT-B]. Developers reuse proven components rather than writing new ones.
4. **Smaller, cheaper, lower-power implementations**: a device implements **only the profiles it needs** (a mouse doesn't need file transfer), keeping Bluetooth's key attributes of **low cost and low power** [BT-B].
5. **Consistent behaviour and easy setup**: every device supports the **Generic Access Profile**, so discovery, connection and security work the same everywhere. With **SDP**, devices can query each other's services and connect automatically. This gives plug-and-play operation and simpler testing and certification [BT-B].

Q8. Scatternet of 2 piconets, no CSMA, 80 frequencies: probability of collision? (10)

Model and assumptions (3 marks):

- Each piconet's master sets a **pseudorandom hop sequence** from its own address, so the two piconets hop **independently** [BT-B].
- **80** equally likely frequencies; one hop per slot.
- Both piconets **always transmit** (always have packets), with **no carrier sensing**.
- A **collision** occurs when both use the **same frequency in the same slot**.
- A **successful transmission** = poll (master → slave) **and** reply (slave → master), so **2 slots on 2 different hops**.

Calculation (5 marks):

1. P(same frequency in a given slot) = 80 × (1/80)(1/80) = **1/80 = 0.0125**
2. P(no collision in a slot) = 1 − 1/80 = **79/80 = 0.9875**
3. The two slots are independent, so P(poll AND reply both collision-free) = $(79/80)^2$ = **0.97516**
4. **P(collision during the transaction) = $1 - (79/80)^2 = 159/6400 \approx 0.0248$**

Answer (2 marks): **about 2.5%** of poll/response transactions suffer a collision (1.25% for any single packet). Performance degrades as more piconets are added, as the notes say: *"as the number of piconets in an area increases, the number of collisions increases"* [BT-B]. SUPPLEMENTARY: with k other piconets, P(slot OK) = $(79/80)^k$.

Q9. A Bluetooth device can be in two piconets at the same time. Is there any reason one device cannot be the controller in both at the same time? (5)

Yes. A device cannot be master of two piconets. A piconet is **defined by its master**: the **frequency-hopping sequence (channel) and the phase (timing)** used by all its members are computed **from the master's own Bluetooth address and clock** [BT-B]. If one device were master of both, **both would follow the identical hopping sequence and timing, i.e. they would be the same piconet**, not two separate ones (and together they would be limited to 7 active slaves). A device **can** belong to two piconets as **slave in both** or **master in one and slave in the other**, time-sharing between the two hop sequences. That is how scatternets are formed [BT-B Fig 12.2].

Also: a device in two piconets must **time-share**. It follows one piconet's hop sequence and timing, then the other's. As master it would have to **impose its own hop sequence on both**, which would merge them into one.

Q10. Two ISPs in a café. (i) Both on channel 11. (10) (ii) Channels 1 and 11. (10)

(i) (10 marks)

- **Does it break down?** No. 802.11's **CSMA/CA is distributed**: every station senses the channel before transmitting, whatever its ISP or AP. The two BSSs simply become **one shared collision domain** on channel 11 [WL 32–35].
- **Normal operation**: when a station of ISP 1 is transmitting, stations of ISP 2 that hear it (or its RTS/CTS) **defer**. Their **NAV** is set from the duration fields, so the channel is **time-shared** between the two BSSs. Each AP only processes frames carrying **its own BSSID**; the other ISP's frames are ignored after reception. The separate IP address blocks are irrelevant at the MAC layer.
- **Simultaneous transmission by two stations** (one per ISP): if both finish DIFS + backoff in the same slot, or they are **hidden** from each other, their frames **collide** at the receiving AP(s). With no collision detection, both frames are lost; **no ACK** returns; each station assumes a collision, **doubles its contention window**, picks a new random backoff and **retransmits** [WL 33, 35, 38].
- **Consequences**: more collisions, retransmissions and backoff delay; the channel's capacity is **shared**, so each ISP gets roughly **half the throughput**; **co-channel interference** raises the error rate

[WL 12; WCP].

(ii) (10 marks)

- Channel 1 (2.412 GHz) and channel 11 (2.462 GHz) are **50 MHz apart**. Each ~22 MHz-wide signal [WC-S 4] does not overlap the other: they are **non-overlapping channels** (SUPPLEMENTARY).
- The two BSSs are now **independent collision domains**. A station's carrier sense on channel 1 does **not** detect transmissions on channel 11, and vice versa.
- **Two stations of different ISPs transmitting at the same time do not collide.** Both frames are received correctly by their own APs and ACKed.
- Each ISP gets the **full capacity** of its own channel, so aggregate throughput roughly **doubles**, with fewer collisions, backoffs and retransmissions and **no co-channel interference** between the networks.
- Stations of the same ISP still contend with each other as normal under CSMA/CA.

Exam Paper, Page 4 — Question 4(e) and Question 5

4(e) Scatternet of 2 piconets, no CSMA. What is the probability of a collision between packets of the two piconets? Successful transmission needs a successful poll (master → slave) and a successful reply (slave → master). 80 frequencies; all piconets always have packets. (4)

Assumptions (from [BT-B]): each piconet hops **independently and pseudorandomly** over the 80 frequencies; both transmit in every slot; a collision occurs when both use the **same frequency in the same slot**.

1. P(collision in one slot) = **1/80**.
2. P(a slot is collision-free) = **79/80**.
3. A transaction = **poll + reply** = 2 slots on 2 independent hops: P(success) = $(79/80)^2$ = 6241/6400 = **0.9752**.
4. **P(collision) = $1 - (79/80)^2$ = 159/6400 ≈ 0.0248 ≈ 2.5%.**

(Per individual packet the probability is 1/80 = 1.25%. Because a successful exchange needs **two** collision-free packets, the transaction collision probability is about **2.5%**.)

5(a) As a mobile node moves farther from a base station, what two actions could the BS take so that the loss probability of a transmitted frame does not increase? (2)

1. **Increase its transmission power**, to compensate for the larger path loss ($\propto d^2$) and keep the SNR up [WC-B 5.2].
2. **Reduce the transmission rate**: switch to a **more robust modulation and coding scheme** (adaptive modulation and coding / 802.11 rate adaptation), which tolerates the lower SNR [WC-S 51; WL 23].

5(b) Use Figure 2 to explain the "hidden node" problem. What happens when A and C start sending simultaneously? (4)

In Figure 2, **B is within range of both A and C, but A and C are outside each other's range**. When A transmits to B, **C cannot hear A**, so C's carrier sense finds the channel idle. A and C are **hidden from each other with respect to B** [FZ]. **If A and C send at the same time** (e.g. both to B): both sense an idle medium and transmit. Their signals **overlap at B, so B receives a collision** and can decode neither frame. Since wireless stations **cannot detect collisions**, A and C keep sending their **whole frames**, wasting the channel. Neither receives an **ACK**, so both time out, **back off a random time and retransmit**. Hidden nodes therefore **reduce network capacity** [FZ; WL 29].

5(c) Use Figure 2 to explain how CSMA/CA is realised in IEEE 802.11. Assume A wants to send a frame to the destination (B). (4)

1. **Carrier sense**: A checks its **NAV** and senses the medium. When idle, it waits **DIFS**, then counts down a **random backoff** (the counter freezes if the medium becomes busy) [WL 32–35].
2. **RTS**: A sends an RTS containing the **duration** of the whole exchange. B hears it (C doesn't).
3. **CTS**: after **SIFS**, B replies with a CTS that also carries the duration. **C is in B's range, so it hears the CTS, sets its NAV and defers**, even though it can't hear A. This is how the hidden node is handled [FZ; WL 39, 43].
4. **DATA and ACK**: after SIFS, A sends the data. After SIFS, B returns an **ACK**. If A gets no CTS or no ACK, it assumes a collision, backs off and retries [WL 35, 38].

5(d) Why are acknowledgements used in 802.11 but not in wired Ethernet? (2)

In wired Ethernet (CSMA/CD) the sender **detects collisions**, and *"if a collision is not detected, it is a kind of acknowledgment that the frame was received"*; wired error rates are also very low [FZ]. In 802.11, a station **cannot detect collisions** (it can't send and receive at once; hidden stations; fading), and frames are frequently lost to **noise, interference and fading**. So *"the station does not have any means to check for the successful arrival of its data"*, and an **explicit ACK** (with retransmission on timeout) is required [FZ; WL 27].

5(e) Two ISPs provide WiFi in a café, each with its own AP and IP address block.

(i) By accident both APs use channel 11. Will 802.11 completely break down? What happens when two stations, each associated with a different ISP, transmit at the same time? (4)

No, it does not break down. The two BSSs simply **share one channel as a single collision domain**. Every station still runs **CSMA/CA**: it senses the channel and its NAV, so stations from **both** ISPs **defer to each other** and take turns, as if they were one bigger BSS. (The different IP blocks don't matter at the MAC layer; each AP only accepts frames addressed to its own BSS.) If two stations from different ISPs **transmit at the same moment** (e.g. their backoff counters expire together, or they are hidden from each other), their frames **collide**. Neither AP receives a correct frame, **no ACKs are returned**, both stations **back off a random time** (with a larger contention window) and **retransmit**. The result is **more collisions and lower throughput for both ISPs** (each gets roughly half the channel capacity), but the network keeps working.

(ii) Now one AP uses channel 1 and the other channel 11. How do the answers change? (4)

Channels **1 and 11 do not overlap** (in the 2.4 GHz band, signals are ~22 MHz wide [WC-S 4] and channels are 5 MHz apart, so 1 and 11 are 50 MHz apart; SUPPLEMENTARY). The two BSSs now form **two independent collision domains**:

- Stations of different ISPs **don't hear each other's transmissions** in carrier sense, so they **don't defer to each other**.
- **Two stations can transmit at the same time without colliding**: each frame reaches its own AP successfully and is ACKed.
- **Each ISP gets the full capacity of its own channel**: aggregate throughput roughly **doubles**, with fewer collisions and backoffs.